

0:00	Title	“Good morning. My name is Martin Maxa, from the Department of Measurement here at this faculty. I worked on this together with Jakub Svatoš, from the same department.” ■ today: detection, classification, localization of impulsive sounds – gunshots as the target class ■ main focus: gunshot classification – cepstral features + two classifiers; localization briefly
0:45	Motivation	“This work is part of the development of an acoustic sensor system for the protection of soft targets.” ■ complements cameras: omnidirectional, no direct line of sight ■ [CUT] complete system: detect, classify (to calibre), localize – outdoors, noise, many reflections ■ [CLICK] three questions: (1) which cepstral representation separates gunshots from similar impulsive sounds, (2) NN or SVM, (3) frame length vs. reflections
1:50	Signature	“Let us first look at the signal of a gunshot. A gunshot has one or two components, depending on the projectile speed.” ■ subsonic → muzzle blast only; supersonic → also a shock wave “this recording is a supersonic shot, so we see both” ■ shock wave ≈ 2.5 ms – [CUT] along the trajectory, N-shaped pressure profile ■ [CLICK] muzzle blast ≈ 45 ms later – gases at the barrel, every gunshot ■ between and after = reflections – “they matter for the choice of frame length, and I will return to them in the results”
3:00	Single-domain	“This also shows why one signal domain alone is not enough.” ■ spectrum → often the environment more than the gunshot ■ time-domain shape → changes during propagation, obstacles ■ ⇒ features combining both domains – cepstral features do exactly that
3:30	System	“Now let us look at the proposed system. It consists of several independent sensor units connected to a remote server.” ■ unit monitors continuously → on an impulsive event, sends the signal ■ server: classification + localization of the source ■ several units: wider coverage + position estimate tested with several calibres and ammunition types
4:20	Localize	“First, I will briefly explain the localization. Each sensor unit has two MEMS microphones, about twenty centimetres apart, sampled at 44.1 kilohertz.” ■ TDoA between the microphones → $\Theta = \arcsin(c \Delta n / (df_s))$ → direction to the source ■ [CLICK] one unit = one direction; two units → intersection = position ■ 70 events, 9 reference angles – cross-corr + parabolic interpolation, MAE 0.77°
5:20	Features	“Now I will move to the main part of the work: classification. We compared four different cepstral feature extractors.” ■ same chain: framing, FFT, filter bank, log, DCT main difference = the filter bank, all four on the slide ■ NEXT (5:50) MFCC mel, dense low (from speech; works for gunshots) LFCC linear IMFCC inverted mel, dense high GTCC gammatone, auditory model, more noise-robust (grid $4 \times 3 \times 2$ is on the slide)
6:25	TABLE → CHART	“This table shows the complete classification results for the nine-millimetre class.” ■ GTCC rows highlighted do not read the numbers → NEXT (6:40) “This chart shows the same results, but only the Matthews correlation coefficient.” ■ MCC: whole confusion matrix, less affected by class imbalance ■ red = GTCC on top in both panels, every frame length NN better than SVM, difference grows with longer frames ■ best GTCC + NN, 50 ms → 99.15 / 96.77 / 0.98 ⇒ 117 of 118 test events correct (31 of them 9 mm; 30 of 31 recalled)
7:35	Discussion	“From these results, we can make three main observations.” ■ MFCC, LFCC, IMFCC similar – filter spacing makes little difference ■ GTCC clearly better, every frame length, both classifiers – [CUT] auditory model, noise-robust: <i>possible</i> reason, not studied directly ■ NN slightly better than SVM, especially longer frames – <i>general trend, not every single result</i> ■ [CLICK] longer frames improve accuracy, <i>but</i> 50 ms already includes more reflections – “so even longer frames would not necessarily help”
8:45	Conclusion	“To conclude.” ■ distributed sensor system → detection, classification, localization ■ cepstral features work well with both classifiers ■ GTCC best; NN slightly better than SVM, especially with longer frames ■ evaluated on real firearms, several calibres and ammunition types
9:30		“Thank you.”

If asked, the numbers. Corpus 371 train / 118 test, five classes (false alarms 86/21, **9 mm 70/31**, 5.56 NATO SD 100/31, 7.62 Tokarev 55/17, .22 60/18) | 26 features per extractor, $\approx 75/25$ split | NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations | bearing: cross-corr 0.84°, +parabolic 0.77°, PBDE 3.24°, PBDE+linreg 3.91° | detection stage = median-filter trigger, RTSI 2019, down to SNR ≈ 5 dB | appendix = last 4 PDF pages: **A1** setup **A2** bearing **A3** other cepstral **A4** metrics (RLC = recall).

Running late? Three passages are marked **[CUT IF LATE]** in the notes – **Motivation** (the three functions), **Signature** (Mach cone), **Discussion** (probable reason GTCC wins). All three ≈ 35 s; dropping them breaks nothing. **Checkpoint:** start the table at **6:25**; past 6:55, cut all three.

Do not guess: 2D position error in metres, SNR limits of *this* classifier, firing distances in this dataset, provenance of the waveform on slide 3. “We did not characterize that in this study – Jakub Svatoš ran the campaign, I can put you in touch.”